

Tutorial Quiz 5 Rubric

Q1]

$$\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$$

Integrate w.r.t. z

$$\begin{aligned} \int_0^1 (x^2 + y^2 + z^2) dz \\ = \left[x^2 z + y^2 z + \frac{z^3}{3} \right]_0^1 \\ = x^2 + y^2 + \frac{1}{3} \end{aligned}$$

Integrate w.r.t. y :

$$\begin{aligned} \int_0^1 (x^2 + y^2 + \frac{1}{3}) dy \\ = \left[x^2 y + \frac{y^3}{3} + \frac{y}{3} \right]_0^1 \\ = x^2 + \frac{1}{3} + \frac{1}{3} = x^2 + \frac{2}{3} \end{aligned}$$

Integrate w.r.t. x :

$$\begin{aligned} \int_0^1 (x^2 + \frac{2}{3}) dx &= \left[\frac{x^3}{3} + \frac{2x}{3} \right]_0^1 \\ &= \frac{1}{3} + \frac{2}{3} = 1 \quad \text{--- (1)} \end{aligned}$$

Q2] Find the volume of the region enclosed by
the cylinder $x^2 + y^2 = 4$
the plane $z = 0$
the plane $y + z = 4$

The region lies above the x - y plane ($z=0$)
and inside the cylinder $r^2=4$

The top surface is the plane $z=4-y$

* In cylindrical coordinates:

$$r = 2, \quad z = 0, \quad \text{and} \quad z = 4 - r \sin \theta$$

$\therefore$ the limits of integration are:

$$\rightarrow 0 \leq r \leq 2$$

$$\rightarrow 0 \leq \theta \leq 2\pi$$

$$\rightarrow 0 \leq z \leq 4 - r \sin \theta$$

So, the final volume integral is:

$$V = \iiint_D dV = \int_0^{2\pi} \int_0^2 \int_0^{4-r\sin\theta} r \, dz \, dr \, d\theta \quad - (1)$$

Now evaluating the integral:

First w.r.t. z

$$\begin{aligned}\int_0^{4-r\sin\theta} r \, dz \\&= r [z]_0^{4-r\sin\theta} \\&= 4r - r^2\sin\theta\end{aligned}$$

Now w.r.t. r

$$\begin{aligned}\int_0^2 (4r - r^2\sin\theta) \, dr \\&= \left[2r^2 - \frac{r^3}{3}\sin\theta \right]_0^2 \\&= 8 - \frac{8}{3}\sin\theta\end{aligned}$$

Now w.r.t. θ

$$\begin{aligned}\int_0^{2\pi} \left(8 - \frac{8}{3}\sin\theta \right) d\theta \\&= \left[8\theta + \frac{8}{3}\cos\theta \right]_0^{2\pi} \\&= \left[16\pi + \frac{8}{3} \right] - \left[0 + \frac{8}{3} \right] \\&= 16\pi \quad \text{--- (1)}\end{aligned}$$

Another way: [Cartesian Coordinates]

z bounds: z ranges from bottom plane to top plane

$$0 \leq z \leq 4 - y$$

The projection of the region in xy plane is the disk $x^2 + y^2 \leq 4$

For a fixed y value where $-2 \leq y \leq 2$

we have
$$-\sqrt{4-y^2} \leq x \leq \sqrt{4-y^2}$$

$\therefore$ the volume integral:

$$V = \int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \int_0^{4-y} dz \, dx \, dy \quad \text{--- (1)}$$

$$= \int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} [z]_0^{4-y} dx \, dy$$

$$= \int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (4-y) dx \, dy$$

$$= \int_{-2}^2 \left[(4-y) \cdot x \right]_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dy$$

$$= \int_{-2}^2 (4-y) \cdot 2\sqrt{4-y^2} \, dy$$

$$= \underbrace{2 \int_{-2}^2 4\sqrt{4-y^2} dy}_{8\pi} - \underbrace{2 \int_{-2}^2 y \sqrt{4-y^2} dy}_0$$

$$= 2 \cdot 8\pi - 2 \cdot (0)$$

$$= 16\pi \quad - \textcircled{1}$$